SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 06 **COURSE NAME:** MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 **Total Marks:25**

Annexure A

Analytical Problem Solving

Code of Conduct:

I M. Ajay (192312542 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M. Ajay

Annexure A

Short Answer Test

1, A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is –0.5. For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails. (5 Marks)

2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training. (5 Marks)

3. A deep learning model is developed for disease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positives, 150 true negatives, 20 false positives, and 50 false negatives. Using these values, compute the accuracy, precision, sensitivity (recall), and specificity of the model. Based on the calculated results, discuss whether the model is reliable for medical diagnosis, considering the importance of minimizing diagnostic errors. (5 Marks)

4.A Genetic Algorithm is applied to maximize the function $f(x) = x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms. (5 Marks)

5 .During the training of a deep neural network model, the training and validation accuracies are observed over multiple epochs. The training accuracy increases steadily from 60% to 95%, while the validation accuracy initially increases from 58% to 72% but then decreases to 68% in later epochs. Analyse this trend and identify the issue affecting the model after a certain point in training. Explain why this issue occurs in neural networks and suggest two effective techniques to address it. Also, discuss the significance of validation accuracy in evaluating the performance of a learning model. (5 Marks)

M.ADAY
1923125142

Perceptron Model- Pass or Fail

Study hours (x_1) = 2

Attendance (x_2) = 1

weights: $w_1 = 0.6$, $w_2 = 0.4$

Bias = 0.5

calculate, Net input

$$\text{Net Input} = x_1w_1 + x_2w_2 + b$$

substitute, the values

$$= (2 \times 0.6) \times ((1 \times 0.4) - 0.5)$$

$$= 1.2 + 0.4 - 0.5$$

$$= 1.1$$

Apply, step activation function

* If net input ≥ 0, output = 1
* If net input < 0, output = 0

since, $1.1 \geq 0$, the output is 1

conclusion: The student passes

(2) Initial weight = 0.5

learning rate(η) = 0.1

Input (x) = 2

Actual output = 0.6

Target output = 1

Calculate Error

Error = Target − Actual

= 1 − 0.6

= 0.4

Calculate weight update using delta rule

$\Delta w = n \times Error \times x$

$= 0.1 \times 0.4 \times 2$

$= 0.08$

Calculate new weight

$w_{new} = w_{old} + \Delta w$

$= 0.5 + 0.08 = 0.58$

The updated weight after one iteration is 0.58

③ Given

True Positive (TP) = 180

True Negative (TN) = 150

False Positive (FP) = 20

False Negative (FN) = 50

Total observations:

$= 180 + 150 + 20 + 50$

$= 400$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \times 100$$

$$= \frac{180 + 150}{400} \times 100$$

$= 82.5\%$

precision = $\frac{TP}{TP+FP} \times 100$

$= \frac{180}{180+020} \times 100$

$= 90\%$

sensitivity = $\frac{TP}{TP+FN} \times 100$

$= \frac{180}{180+50} \times 100$

$= 78.26\%$

specificity = $\frac{TN}{TN+FP} \times 100$

$= \frac{150}{150+20} \times 100$

$= 88.24\%$

④ Given $f(x) = x^2$, population = 2, 4, 6, 8

Fitness calculation:

Individual (x)	Fitness x^2
2	4
4	16
6	36
8	64
10	100

Selection: The best individuals are 8 and 6 because they have the highest fitness values (64 and 36)

Crossover: Binary values are 8 = 1000, and 6 = 0110, exchange the right most bits to generate offspring

Mutation: Mutation randomly changes bits in an individual. It maintains diversity, explores new solutions and helps avoid local optima.

Conclusion:
Genetic Algorithms use selection, cross over and mutation to find better solutions

⑤ Problem: The model suffers from overfitting because training accuracy increases to 95% but validation accuracy decreases from 72% to 68%.

Reason: The model learns the training data too well including noise and performs poorly on new data.

Techniques to prevent overfitting:

Dropout: Randomly disables some neurons during training

Early stopping: Stops training when validation performance starts decreasing

Importance of Validation Accuracy:

Measures performance on unseen data
Helps identify overfitting
Helps select the best model

Conclusion:
Dropout and early stopping help reduce overfitting and improve performance on new data.